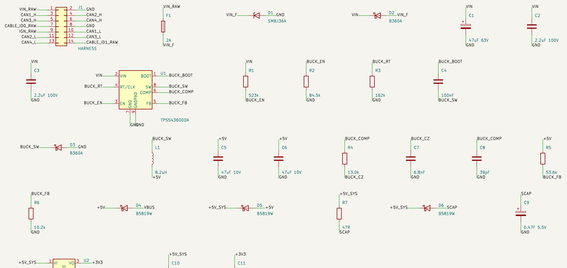
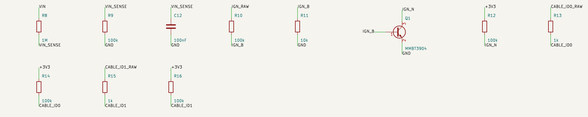
POWER: 9–36V → fuse → TVS → reverse diode → TPS54360 5V buck → hold-up → 3.3V LDO



J1: Harness: 1 VIN, 2/8 GND, 3–6 CAN1–4 H, 10–13 CAN1–4 L, 9 IGN, 7/14 cable ID
D1: TVS: load dump / jump start; reverse polarity blows F1 via D1
D2: Reverse polarity block
U1: Values from TI TPS54360 datasheet 5V design example (Fig. 34)
R2: UVLO: start 8V, stop 6.25V
R3: fsw ~600 kHz
D3: Catch diode
D4: USB can power the whole board for bench work
D5: Isolates hold-up rail from buck/transceivers
R7: Supercap charge limit (~100 mA)
D6: Supercap discharge path
C9: ~4 s hold-up at 0.5 W; lets firmware close log files on power loss

VIN / IGNITION / CABLE-ID SENSE



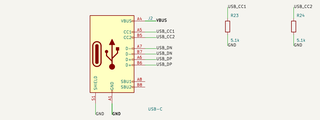
R9: VIN/11 → ADC: 36V = 3.27V
Q1: IGN > ~6.5V pulls IGN_N low (9–60V safe)
R16: Cable grounds ID0/ID1 to identify fuelf

MCU: ESP32-C5-WROOM-1 (2× native CAN-FD, Wi-Fi 6 + Wi-Fi Aware)

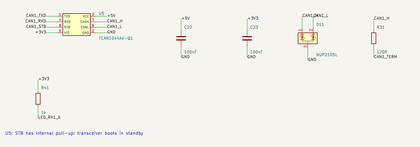


U3: GPIO28 = I2C SCL + BOOT strap (pull-up = SPI boot)
SW2: Hold BOOT while tapping RESET for download mode

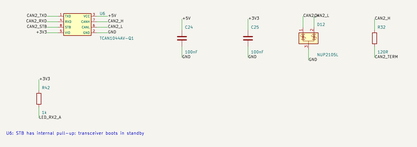
USB-C (native USB Serial/JTAG)



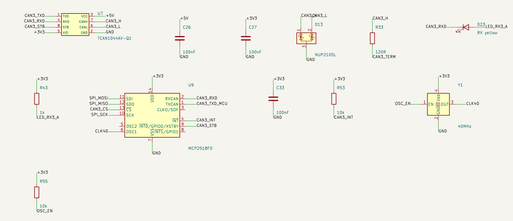
CAN1 (native CAN-FD 0) — OBD gateway pins 6/14



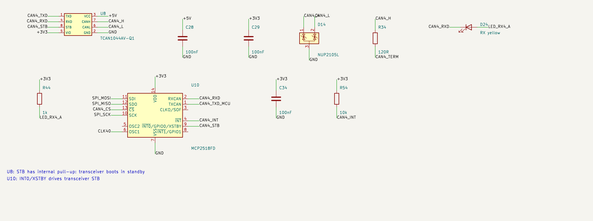
CAN2 (native CAN-FD 1)



CAN3 (MCP2518FD #1) + shared 40MHz clock



CAN4 (MCP2518FD #2)



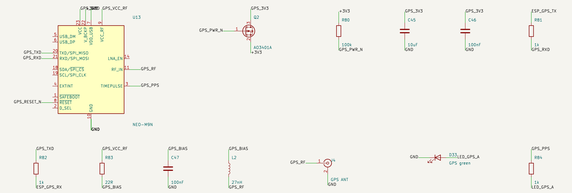
TX-ENABLE and TERMINATION jumpers

microSD (SPI)

IO EXPANDER, STATUS LEDS, MARK BUTTON

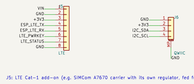
RTC + CR1220 backup

GPS (optional) NEO-M9N 25Hz + active antenna bias-T



U13: Optional; leave U13 + bias-T unpopulated to rely on the phone's GPS
Q2: GPS power switch
R63: Bias-T per NEO-M9N integration manual; limits short-circuit current
R64: Blinks once per second with a fix

EXPANSION: LTE header, Qwiic



J5: LTE Cat-1 add-on (e.g. SIMCom A7670 carrier with its own regulator, fed from VIN)

MECHANICAL



FLAGS



All transceivers boot in STANDBY; TX needs firmware + a TX_EN jumper
Generated from gen/design.py — edit that, then run gen/build.sh

Sheet: /	
File: quadcan-c5.kicad_sch	
Title: QuadCAN-C5: 4-channel CAN-FD logger	
Size: A2	Date: 2026-11-12
KiCad 8.0.4	kicad-cli 7.0.11+dfsg-1build4
	Rev: 1.0
	ID: 1/1